Example report for the practical assignment Automated Reasoning 2IMF25

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Several points of attention are mentioned in the green boxes below.

Problem: eight queens problem

We are given a chess board with two queens on it. One is placed on a1 and another on f4. Find a way to put six more queens on the board such that no two queens share a column, row or diagonal.

Solution:

We identify several parameters in our problem:

Provide an overview of the parameters of the problem.

- $n = 8$ is the size of the board and the total number of queens.
- $k = 2$ is the number of queens that is already present.
- x_i and y_i for $i \in \{1, \ldots, k\}$ represent the x and y-coordinate respectively of queens that are already present.

To place our n queens, we introduce the following variables:

Indicate the type of variables

- $p_{ij} \in \mathbb{B}$ for $i, j = 1, \ldots, n$: the value of p_{ij} will be true if and only if a queen will be put on position (i, j), that is, in the i-th row and in the j-th column

Setting the location of already-placed queens is done by setting the corresponding location to *true*

$$\bigwedge_{i=1}^{k} p_{x_i, y_i}$$

As we have to put exactly n queens, and no two are allowed to be on the same row, every

Present formulas compactly, making it easy to understand the structure.

row should contain at least one queen. For row i this is expressed by the formula

$$\bigvee_{j=1}^{n} p_{ij}.$$

In a similar way every column should contain at least one queen; for column j this is expressed by the formula

$$\bigvee_{i=1}^{n} p_{ij}.$$

Next we express that every row contains at most one queen, that is, for every two distinct positions j, k it is not allowed that both p_{ij} and p_{ik} are true. For row i this is expressed by the formula

$$\bigwedge_{j,k:1\leq j<k\leq n} \neg p_{ij} \vee \neg p_{ik}.$$

Similarly, every column should contain at most one queen; for column j this is expressed by the formula

$$\bigwedge_{i,k:1\leq i<k\leq n} \neg p_{ij} \vee \neg p_{kj}$$

Finally, we consider the requirements on diagonals. Two positions (i, j) and (k, m) are on the same diagonal in the one direction if and only if $i + k = j + m$, and they are on the same diagonal in the other direction if and only if $i - k = j - m$. For every pair of such positions it is not allowed that they are both occupied by a queen, so we require

Include parentheses when necessary

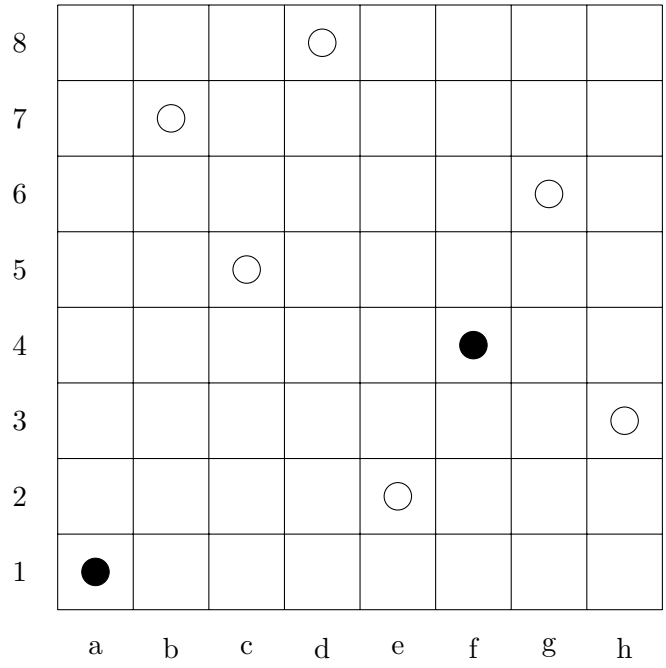
$$\left(\bigwedge_{\substack{1\leq i,j,k,m\leq n \\ i+k=j+m}} \neg p_{ij} \vee \neg p_{km}\right) \wedge \left(\bigwedge_{\substack{1\leq i,j,k,m\leq n \\ i-k=j-m}} \neg p_{ij} \vee \neg p_{km}\right)$$

The total formula now consists of the conjunction of the constraints above.

This SAT formula is easily constructed using the Python API of Z3. Running the solver yields a solution in a fraction of a second.

Report on approach and the running time

In our solution, the values that are are true are $p_{14}, p_{22}, p_{37}, p_{43}, p_{56}, p_{68}, p_{75}, p_{81}$. This yields the following board, where pre-placed queens are coloured black:



A clear visualisation showing the solution

We check that indeed there are eight queens for which no two are on the same row, column or diagonal.

Remark:

Our formula contains some redundancy: the requirement that every row contains exactly one queen implies that there are exactly n queens in total. By expressing that every column contains at least one queen, from this property one concludes that also every column contains at most one queen. We chose for this redundancy for keeping the symmetry in the solution, and also following the general strategy that introducing redundancy is often good for efficiency.